

Pradnya Chandrakant Bhakare

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London, UK | GWR Visa Eligible

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| <https://pradnya13-ts.github.io/>

EDUCATION

University	Degree	Year	Grade
Queen Mary University of London, (UK)	MSc, Artificial Intelligence and Machine Learning in Science	2025-2026	In Progress
Sharnbasva University, Kalaburagi (India)	Bachelor's of Computer Applications	2022-2025	9.23 CGPA

PERSONAL PROFILE

A technically proficient MSc AI & Machine Learning student from Queen Mary University of London with a strong foundation in software engineering. Skilled in designing and deploying end-to-end ML systems and agentic AI workflows, from algorithm development in Python (PyTorch/TensorFlow) to LLM integration via the Model Context Protocol (MCP). Passionate about MLOps, agentic AI, and emerging patterns like Karpathy's LLM Wiki, and actively engaged with the local technical community through GDG London.

WORK EXPERIENCE

DeepTrace AI | Agentic AI & MLOps Engineer

May 2026 - Present

London, UK

- AI Orchestration Platform:** Architected and deployed an AI orchestration layer coordinating multiple specialized AI agents, enabling reliable task routing, context sharing, and tool-use across distributed agent workflows.
- Multi-Agent System Design:** Designed and built multiple autonomous AI agents with distinct roles (analysis, retrieval, decision-making), integrating them into a unified pipeline for end-to-end automation of complex tasks.
- Deepfake Detection Platform:** Architected and deployed the initial prototype for a deepfake detection platform, translating conceptual AI research into a functional, scalable product.
- ML Pipeline Design:** Designed and optimized advanced machine learning pipelines using computer vision and sequence models to detect digital manipulations with high precision.

Queen Mary University of London | Research Assistant, Quantum ML

Feb 2026 - May 2026

London, UK

- Quantum ML Workflows:** Developed and implemented quantum machine learning workflows to simulate large metalloenzymes and evaluate electronic structures for protein-ligand binding.
- Molecular Simulation:** Leveraged Density Functional Theory (DFT) data and advanced Python chemistry libraries to build predictive machine learning loops for molecular simulation.

Digisnare Technologies LLP | AI and ML Intern

April 13 - July 13 2025

Kalaburagi, India

- Model Development & Impact:** Developed and fine-tuned a machine learning model (Gradient Boosting) that achieved 90% prediction accuracy on a real-world task, exceeding the previous baseline performance by 15%.
- Problem Solving & Application:** Applied Time Series Analysis and Clustering to solve a core business challenge, resulting in a 20% reduction in system downtime and a 10% efficiency gain in resource allocation.

PROJECTS

Prompt2Pixel — Cloud-Native Text-to-Image Generation Powered by Stable Diffusion and Google Cloud - [Github](#)

- Built and deployed a containerized text-to-image generation API on Google Cloud Platform, combining FastAPI, Stable Diffusion v1.5, and Cloud Storage for end-to-end image generation, while managing GCP infrastructure and Docker deployment.

Martian Surface Feature Classifier - CNN - [GitHub](#)

- Designed a custom 3-block CNN trained on NASA HiRISE orbital imagery to automatically classify 8 Martian geological surface features including craters, dunes, and spider formations, achieving 90.85% test accuracy.

Personal Knowledge Base using the LLM Wiki Pattern (Karpathy)

- Implemented Karpathy's LLM Wiki pattern using an AI coding agent to compile raw research into structured, interlinked markdown pages, building a self-maintaining knowledge base that updates and self-audits over a traditional RAG pipeline.

Hybrid SVM for Cryptocurrency Market Forecasting - [GitHub](#)

- Designed and implemented a Hybrid Support Vector Machine (SVR/SVC) model using Python/Scikit-learn for cryptocurrency forecasting, achieving competitive RMSE for price prediction and over 60% accuracy in directional trend classification.

Jarvis - AI Voice Assistant for Smart Home - [GitHub](#)

- Built a Python-based voice assistant using NLTK/spaCy supporting 10+ task automation commands, integrated with 3 IoT device models for smart home control.

Vita AI - Disease Prediction System - [GitHub](#)

- Built an ML-based system using 5 classification algorithms, including Logistic Regression and Random Forest, to predict 8 common diseases from user-input symptoms, achieving an average F1-score of 87.5% across all classes.

Object Detection System for Blind People - [GitHub](#)

- Designed and trained a CNN/YOLOvX object detection model using TensorFlow/Keras and OpenCV to classify 20+ real-world objects, providing real-time audio feedback.

ACHIEVEMENTS AND CERTIFICATIONS

Wins: 1st Place, [IBM Hackathon](#) | 1st Prize, Technical Quiz Championship organized by [IBM](#).
Certifications: Introduction to Model Context Protocol (MCP) by [Anthropic](#) (hands-on MCP server-client architecture) | Introduction to Kubernetes - [Linux Foundation](#) (edX), 2026 | [Google Certification](#): Data Processing | [IBM Certification](#): Machine Learning with Python | [Deloitte Australia](#) - Data Analytics Job Simulation | [AWS Artificial Intelligence Practitioner](#) Learning Plan

TECHNICAL SKILLS

Languages: Python (Advanced), Java, C++, JavaScript
ML / DL: TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, NLP (NLTK/spaCy), OpenCV
Data & DB: Pandas, Seaborn, Matplotlib, SQL, MongoDB
MLOps / Tools: Git/GitHub, Docker, API Integration, Data Structures & Algorithms, Linux, Jupyter
Cloud: Google Cloud (GCP), AWS, Azure
AI Protocols: Model Context Protocol ([MCP](#)), Large Language Model (LLM) Integration, [LLM Wiki](#), Spec development